

Jenkins Distributed Build Lab - Complete Runbook

1. Objective

Deploy a Jenkins controller and SSH agent on separate EC2 instances to understand distributed builds, node scheduling, workspaces, remoting, and troubleshoot Disk Space Monitor warnings.

2. Lab Architecture

```
GitHub
|
Jenkins Controller (EC2 t3.small)
|
SSH (Launch agent)
|
Linux Agent (EC2 t3.small)
```

3. Prerequisites

Two Ubuntu EC2 t3.small instances, security groups allowing SSH (22) and Jenkins (8080 if needed), Java installed on both, Jenkins installed on controller.

4. Configure SSH Agent

```
# On agent
sudo apt update
sudo apt install -y openjdk-21-jre

mkdir -p ~/jenkins-agent
```

In Jenkins: Manage Jenkins -> Nodes -> New Node -> Permanent Agent -> Launch agents via SSH.

5. Verify Agent

```
ps -ef | grep remoting
# Expected
java -jar remoting.jar -workDir /home/ubuntu/jenkins-agent
```

6. Verify Scheduling

```
pipeline {
  agent { label 'linux-agent' }
  stages {
    stage('Verify') {
      steps {
        sh 'hostname'
        sh 'pwd'
      }
    }
  }
}
```

7. Investigate Disk Warning

```
df -h /tmp
# showed tmpfs ~953M

# Script Console
println System.getProperty("java.io.tmpdir")
# /tmp
```

Root filesystem had ~3.6GB free, but Jenkins warned because java.io.tmpdir pointed to /tmp (tmpfs) whose total capacity was below the 1GiB threshold.

8. Controller Fix

```
sudo mkdir -p /var/lib/jenkins/tmp
sudo chown jenkins:jenkins /var/lib/jenkins/tmp
sudo chmod 1777 /var/lib/jenkins/tmp
```

```

sudo systemctl edit jenkins

[Service]
Environment="JENKINS_JAVA_OPTIONS=-Djava.io.tmpdir=/var/lib/jenkins/tmp"

sudo systemctl daemon-reload
sudo systemctl restart jenkins

# Verify
# Script Console:
println System.getProperty("java.io.tmpdir")

```

9. Agent Fix (SSH Build Agents Plugin 3.1097)

```

mkdir -p /home/ubuntu/jenkins-agent/tmp
chmod 1777 /home/ubuntu/jenkins-agent/tmp

sudo tee /usr/local/bin/java-jenkins >/dev/null <<'EOF'
#!/bin/bash
exec /usr/bin/java -Djava.io.tmpdir=/home/ubuntu/jenkins-agent/tmp "$@"
EOF

sudo chmod +x /usr/local/bin/java-jenkins

# Jenkins Node Configuration
# JavaPath:
/usr/local/bin/java-jenkins

# Reconnect agent

ps -ef | grep remoting
# Expected:
# /usr/bin/java -Djava.io.tmpdir=/home/ubuntu/jenkins-agent/tmp -jar remoting.jar ...

```

10. Validation

```

# Controller Script Console
println System.getProperty("java.io.tmpdir")

# Agent process
ps -ef | grep remoting

# Pipeline
node('linux-agent'){
  sh 'hostname'
  sh 'pwd'
}

```

11. Lessons Learned

Controller and agents are independent JVMs. `java.io.tmpdir` is JVM-specific. Ubuntu tmpfs on `/tmp` can trigger false Disk Space Monitor warnings. SSH Build Agents Plugin 3.1097 uses `JavaPath` wrapper to inject JVM options.

12. Cleanup

```

Terminate both EC2 instances.
Delete temporary security groups if no longer required.
Retain this runbook for future recreation.

```